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Drug Pills Identification System using Google Gemini LLM: A Generative AI approach

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***Abstract*—Generative AI is emerging as a disruptive force in the healthcare industry, bringing novel solutions ranging from drugdevelopment and clinical decision support to personalized patient care. This study is focused on drug discovery using the Generative AI model. In this paper, a system is proposed for providing drug descriptions from drug pill images. The system is implemented by utilizing Large Language Models (LLMs) in combination with computer vision to detect and provide detailed information about drugs from pill images. In the proposed system, the identification process begins by taking the medicinal drug pills and their cover images. Then, the image is converted into binary values using a standard built-in function. In addition, the target language for providing audio descriptions about the drugs is also used. Then, theGoogle Gemini LLM model is customized by using binary values of the image, target language, and ontology-based prompt engineering. As a result, the LLM model provides drug descriptions in text. Then, the textual description of the drug is converted into the target language audio format by using the Google Text to Speech Converter. The system is experimented by using 807 medicinal drug images which are collected from web resources. The performance of the system is measured by using accuracy. The system achieved an accuracy of 95.04% which is a little higher when compared with the current state-of-the-art model.**

Keywords*— *Drug Pill Identification, Generative AI, Large Language Models, Pharmaceutical Image Analysis, Multimodal Learning, Medical Text Analysis.

I. Introduction

Smart healthcare is more vital in today's digital environment. Internet of Things (IoT) devices are frequently used in smart healthcare systems [1, 2]. In addition to hardware devices, a software system is required for smart healthcare systems. Specifically, correct identification and description of pharmaceutical medications from pill photographs is critical in a variety of industries, including healthcare, pharmacy management, and regulatory compliance. Mislabeled or counterfeit pharmaceuticals offer substantial health concerns to patients, whilemisidentification can result in dangerous prescription errors in clinical settings. Traditionally, drug identification has relied mainly on manual approaches such as matching pill features (e.g., color, shape, and imprint) to reference databases or packaging specifications. While these approaches work well for small-scale applications, they are labor-intensive, time-consuming, and prone to error, especially when working with pills with unclear impressions

or optical distortions caused by illumination or occlusion. To address these issues, new computer models that use both visual and contextual information are required to generate accurate medication descriptions.

This article presents a unique approach that uses a Large Language Model (LLM) to generate automated pharmacological descriptions from pill photos. The inclusion of LLMs in this assignment is prompted by their high ability to comprehend multimodal inputs as well as their ability to absorb external knowledge about drug properties and applications. By combining visual feature extraction with LLMs' contextual capabilities, the system not only correctly identifies the drug but also generates comprehensive and human-readable descriptions that include critical information such as drug name, dosage, manufacturer, potential uses, and safety warnings.

The proposed system processes pill images along with cover or packaging images, which provide additional visual cues such as brand logos, regulatory information, and dosage details. Using a multimodal LLM, the system generates a unified representation that combines these visual features with relevant textual data, enabling it to produce detailed descriptions that go beyond simple identification.

The primary contributions of this research are as follows:

A. To develop a multimodal drug description generation framework

that leverages LLMs for processing pill images and packaging information to produce detailed and contextually relevant descriptions.

B. To evaluate the system's performance

against conventional image-based and text-based methods in terms of accuracy, completeness, and clarity of the generated descriptions.

The paper is organized as follows: Section 2 reviews related work on drug identification systems. Section 3 presents the background study of LLM and GTTS and section 4 gives the architecture of the proposed system. Section 5 presents experimental results, including performance comparisons with traditional method. Finally, Section 6 concludes the paper and outlines future research directions.

II. Related Works

In the survey [3], the advancements and challenges in utilizing deep generative models (DGMs) for the novo drug design are explored. The Integration of Artificial Intelligence (AI) and computational power, the role of machine learning (ML) and deep learning (DL) in drug discovery, and the potential of DGMs to optimize the drug development process and uncover new therapeutic avenues are also discussed. In addition, the challenges associated with DGM implementation, including data quality, model interpretability, and computational complexity are highlighted.

Deep Convolution Network (DCN) based pill identification is implemented in the work [4]. The DCN model is trained on a dataset of pill images captured under unconstrained environments, such as various angles and illumination conditions and also includes a variety of pill shapes, colors, and imprints. The DCN model is implemented using the Caffe deep learning framework. However, this DCN model requires a large dataset of pill images for training. The model may not perform well on pillimages that are significantly different from the images in the training dataset.

The study [5] demonstrated the potential of NLP models for automated ADR detection in a real-world setting. The model created discharge summary formatting patterns that enable accurate classification of distant relations as long as they are provided within the discharge summary's expected structure. However, one of the limitations ofthis model is that the model was trained on a relatively smalldataset of discharge summaries. This may limit the generalizability of the model to other settings.

The YOLOv8 framework is used in a new mobile application that makes use of a standard smartphone camera to enable real-time pill detection and identification [6]. The program uses a dataset designed especially for pill recognition and optimizes the model through transfer learning in order to get high accuracy in real-world situations. Additionally, the application's incorporation of TTS technologies offers an interactive experience that enables users to obtain instantaneous aural information regarding the pills identified.

In order to democratize database access for non-technical users in the realm of pharmacovigilance (PV), this work presents an application of large language models (LLMs) [7]. The researchers created a chatbot that uses natural language to generate structured query language (SQL) questions using OpenAI's GPT-4. The proposed application is implemented using a retrieval-augmented generation (RAG) framework, which combines LLMs with a business context document. By incorporating the business context document.

the LLM is able to generate more accurate and domain-specific SQL queries.

The potential of artificial intelligence (AI) to revolutionize drug discovery and development is discussed [8]. The challenges, opportunities, and strategies associated with AI implementation in this field are discussed. The authors highlighted the need for collaboration between AI researchers and pharmaceutical companies to overcome data privacy concerns and the requirement for substantial labeled datasets. They also emphasize the importance of ethical considerations and potential job displacement due to automation.

A system is developed to assist visually impaired chronic patients in managing their medication safely and effectively [9]. The system contains a deep learning training server, a mobile app, an intelligent medicine recognition device, and a cloud-based management platform. The system utilizes image processing techniques to identify and classify pill medications based on their size, shape, and color. This information is then stored in a local database and used for future recognition.

In the study, human interactions and driving innovations in the field are utilized in the LLM model. However, a smalldisadvantage in using LLM in this model is finding specialized languages which often lead to imbalance in data. To overcome those difficulties, they used a few methods like pharmBERT, two-stage fine-tuning, and only fine-tuning LayerNorm [10]. The significance of reproducibility in scientific research is discussed, along with how LLMs affectmedical education by encouraging critical thinking instead of memorization and their ability to help with the creation ofscientific texts and the summarization of scientific ideas[11].

Deep learning techniques have revolutionized identification classifiers across a range of domains. Understanding how human identification confusion of look-alike images occurs via the cognitive counterpart of deep learning solutions and, consequently, suggesting additional approaches is the aim of the study [12], which uses a baseline deep learning drug identification (DLDI) to find better image-based solutions for the blister package identification problem.

Drug-drug interactions (DDIs) frequently result in unanticipated adverse effects. The clinical detection of DDIsis critical for both patient safety and healthcare cost containment. Although text-mining-based systems investigate several strategies for classifying DDIs, their classification performance in long and complicated phrases remains inadequate [13].

In the study [14], a pill identification approach based on imprinted characters, with the notion that the most important information in a pill image is contained inside those characters is discussed. A character-level language model is used to accurately identify medications, unlike traditional methods that rely solely on picture classification algorithms.

The development of Generative artificial intelligence (GAI) has a tremendous impact on the identification of novel medicines.

The current state of GAI models and their possible use in drug development were examined in the study [15]. It's possible that the molecular generators based on basic CNNs, GNNs, and RNNs are more effective. A variation inference method is used by VAE, GAN, and related techniques. In contrast to variation inference, RL enables real-time process modification to generate the desired molecules.

The study [16] described a ground-breaking multi-layered architecture for network medicine that employs Generative AI to expedite drug discovery and development. Using medication-target co-occurrences from the literature, the framework identifies potential drug combinations for complex illnesses, with a focus on breast cancer therapy. The framework employs ChatGPT for fast engineering to extract drug names and find potential combinations that have been mentioned in clinical studies.

According to the survey [17], generative AI is crucial to the healthcare sector in areas such as drug discovery, medical imaging, clinical trial optimization, medical simulation and training, mental health support, healthcare operations, and resource management. Chatbots, text generation, summarization, and other uses the technology's adaptability and reliability in the medical domain. Large Language Modelling, a generative AI technique, is employed in this work to extract text from drug pill photos in order to provide drug descriptions. This is an experiment using the Google Gemini LLM model.

III. BACKGROUND STUDY

A. *Large Language Model*

Large Language Models (LLMs) are deep learning models that process and produce human-like text. They are often built on Transformer architectures, which employ self-attention processes to record intricate links between words and phrases in big-text corpora. The key components ofLLMs are**:** i) Model Architecture and ii) Scale and Training Data.

Model Structure: The Transformer model is the main component of the Large Language Model (LLM) architecture and serves as the basis for the majority of cutting-edge LLMs, including Google's Gemini, BERT, and GPT. Transformers are made up of feed-forward networks and attention layers that enable parallel processing and effective long-range dependency capture. Transformers capture context and dependencies over lengthy text sequences by using self-attention to determine each word's relative relevance to the others in the sequence. Transformers employ positional encoding to incorporate the order of words in a sequence because, unlike RNNs, they lack intrinsic sequential processing. In Figure 1, a simple LLM model is displayed.

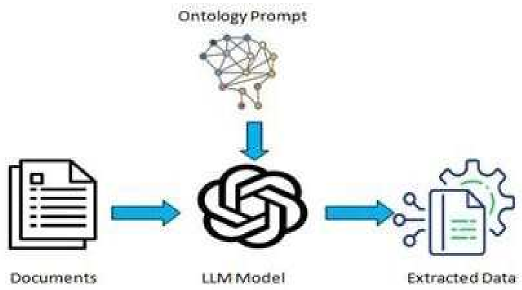

Fig. 1. Basic LLM Model

Scale and Training Data**:**

The vast number of parameters, which can occasionally surpass billions, characterizes LLMs. The latest models, such as GPT-4, have even more parameters than GPT-3, which has 175 billion. To acquire the subtleties of language, including syntax, factual knowledge, and even certain common thinking processes, these models are trained on a variety of text corpora, including books, papers, webpages, and other sources. After pre-training, LLMs can be refined on particular datasets to focus on different applications, like sentiment analysis, chatbots, and summarization.

B. *Google Gemini*

Google's Gemini AI is a new generation of multimodal large language models designed to compete with OpenAI's models like GPT-4. It integrates text, images, audio, and other data types, which allows for more natural and context- aware interactions across various modes of communication. Gemini can process text, images, and other forms of data simultaneously, making it more flexible for varied applications, like advanced conversational agents and cross-media content analysis. There are different versions of Gemini AI which include Gemini 1.0 Ultra, Gemini 1.5, Gemma models, CodeGemma. In this work, the Gemini 1.5 version is utilized.

IV. METHODOLOGY

The proposed system is implemented in three substantial processes that include i) Data Collection ii) Large Language Modelling iii) Text to Audio conversion. Initially, drug pill images are collected from web resources. Then, the LLM is done by using Google Gemini. After which, the textual description of drug information is converted into Audio in the desired language using Google text to speech (GTTS). The proposed system functional diagram is depicted in Fig. 2.

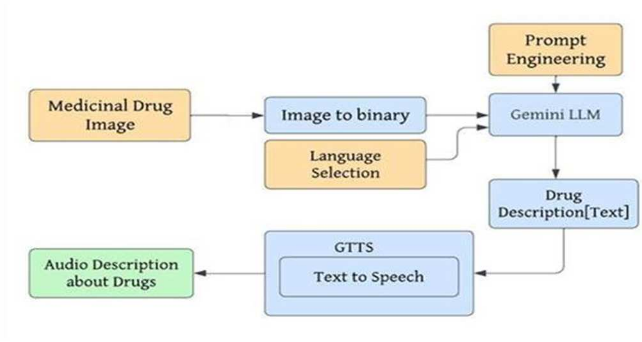

Fig.2.Outline of Proposed System

A. Data Collection

We obtained two separate image data sets from the website Kaggle: i) Medicine tablet pack and ii) Drug name detection. The two datasets are then combined into a single dataset to produce a huge pool of photos for the purpose. As a result, the dataset includes 807 drug pictures. Fig. 3 shows some of the drug images from the dataset.

These images are transformed into binary values using the standard Python library's built-in function. The binary values of the image, the selected language for audio and text,and the prompt text to tailor the LLM are then fed intoGemini. As a result, three features are provided as input to the Gemini LLM.

Fig.3. Sample Medicinal Drug Images in the data set

B. Large Language Modelling using Gemini

This model makes use of the Gemini 1.5 LLM for experimentation. Gemini can process and integrate.

information from a variety of sources; including text, graphics, and audio, making it particularly useful for applications that require multiple inputs. This multimodal method provides a more comprehensive contextual understanding than models like OpenAI's GPT-4, which are primarily text-based.

The model accepts binary picture values, target language, and prompt text as input. Gemini's built-in features will deliver drug descriptions with prompt wording. Fig. 4 depicts the text provided as input in the form of prompt engineering.

```
""" You are an expert in finding the names of medicines or pills. You
will be given images of medicine or pills.
You will have to answer the name of the medicine and the problems it
cures with its uses. Reply in {selected_language}.
If you find that the image is not clear or not a medicine, then provide a
warning: "Recheck your medicine and upload."
```

Fig.4. The textual data in Prompt Engineering

C. Text to Audio conversion

After receiving response from the Gemini model, it is transformed to audio. GTTS is employed to carry out this task. GTTS requires three parameters They are text, lang and slow. The Gemini model's response and chosenlanguage are supplied to GTTS as input, and the output is in audio format. The slow parameter accepts a boolean value and controls the speed of speaking. The audio is then shown in the user interface.

V. Results and Discussion

The system is designed by using Google Gemini 1.5 and GTTS. For drug name identification and providing text descriptions about the drugs Gemini LLM is used. After which, the textual data is converted into speech by utilizing GTTS.

A. UI of the system

In this system, the user will upload an image and select the preferred language. The input is passed along with the prompt to the Gemini 1.5 model and the response is obtained. Also, the response is converted to speech using GTTS and audio is displayed. Correctly predicted text from the drug image and its audio description obtained from the experiments is shown in Fig. 5.

Even when the image is rotated the model can extract thename of the pill and describe the image as shown in Fig. 6.

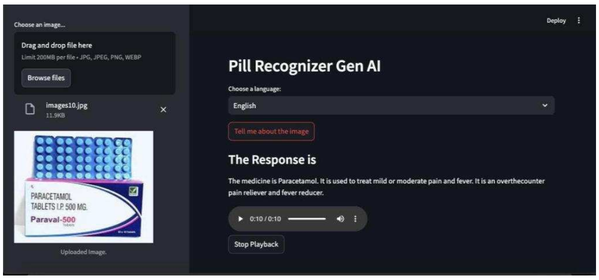

Fig.5. Result of Correctly Predicted Drug Pill and its Audio Description

Fig.6. Result of Correctly Predicted Drug Pill for the Rotated Image and its Audio Description

B. Limitations

There are certain limits to this system. If the image ofthe pill is unclear or does not have a name, the model will be unable to describe the image. Fig. 7 illustrates several photos that the model will not process.

TABLE I. System Performance using Gemini LLM

Statistics	Google GeminiModel	BERT Model
Total Number of Drug Images		8 0 7
Number of Correctly Predicted Images	767	742
Number of Non-Predicted Images	40	65
Accuracy	95.04%	91.94%

Fig.7. Non Predicted Sample Image

C. System Performance

The Gemini model used for pill recognition is evaluated on a dataset comprising 807 drug images. The accuracy of the model is measured by using Eqn 1. For the same dataset we have used the Bidirectional Encoder Representations from Transformers (BERT) model to assess the performance The statistics of the modal performance of GoogleGemini and BERT are given in Table 1.

$$Accuracy = \frac{Total\ number\ of\ pill\ images}{Total\ correct\ predictions} X\ 100$$

Some images with rotated text were not identified in the BERT model, hence the model achieved 91.94%. Because the BERT model recognizes sequential text data and understands natural language based on word order. The proposed Google Gemini Model in recognizing drug pill images performance is higher than the existing state of art which is shown in Fig 7.

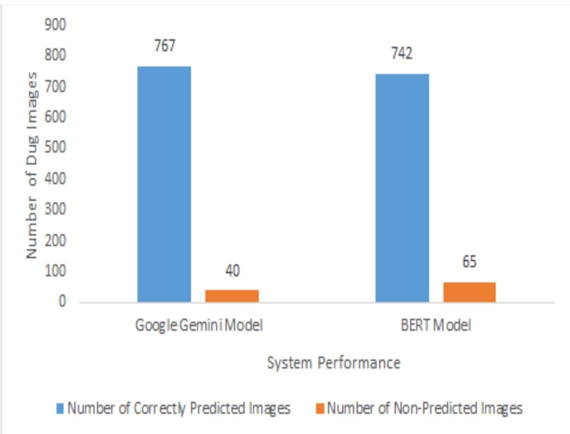

Fig.8. Proposed System Performance

VI. Conclusion

The system developed a smart healthcare solution using the Streamlit application. The system accepts drug images, analyzes them, and provides detailed descriptions of the medicine in Tamil or English, based on the user's choice. The text is then converted to audio using the GTTS service, helping people who have difficulty in reading. The system is experimented by using 807 medicinal drugs images which is collected from web resources. The system performance is assessed by using accuracy and achieved 95.04%. The performance of the model is also compared with other existing state-of-the-art model and proved that the Google Gemini LLM model achieved a little higher performance. Theadvantage of the model is that it can identify the drug name even if the drug cover image is rotated. However, the drug name is not identified if the image and text are not clear.

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